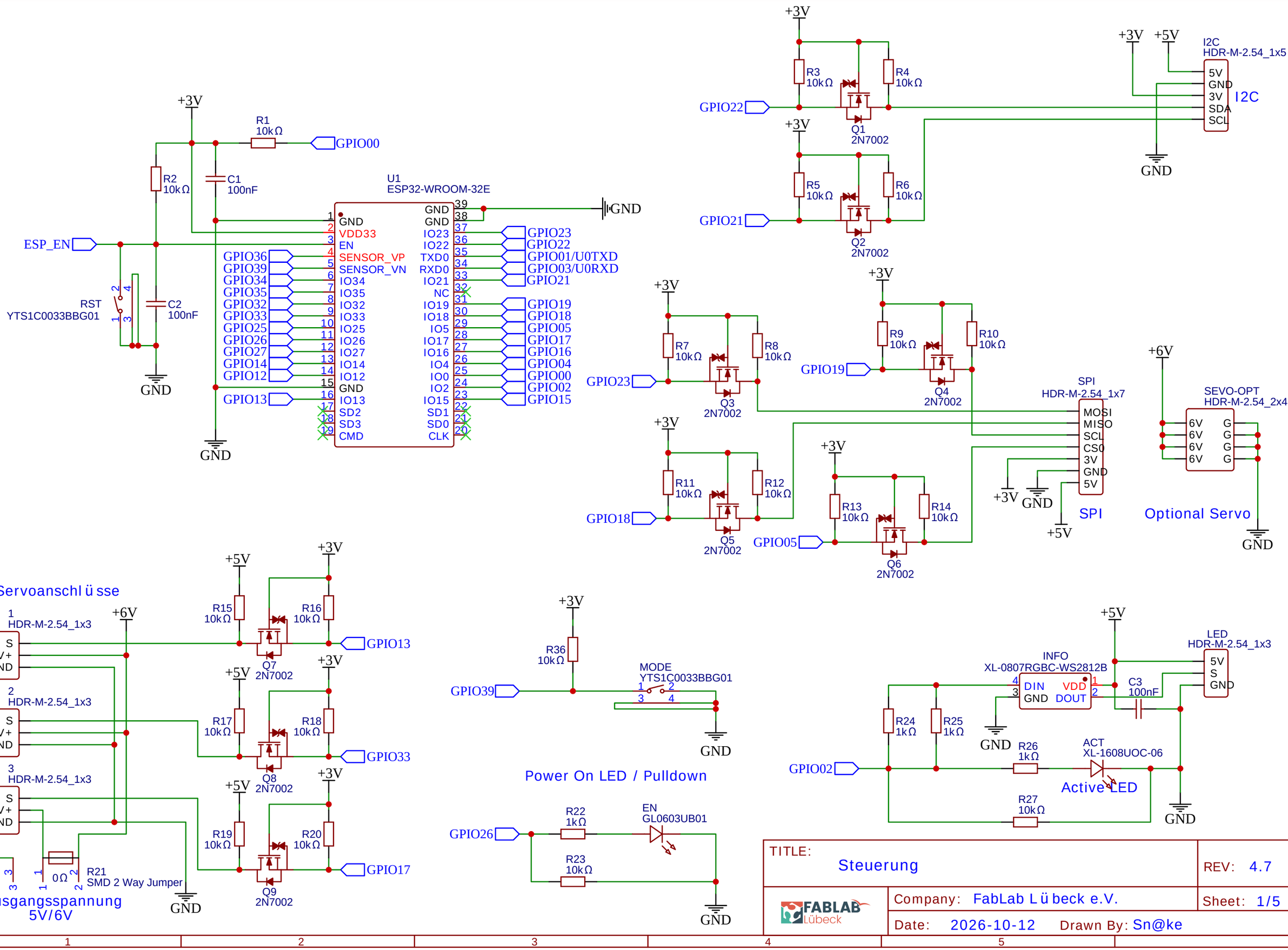
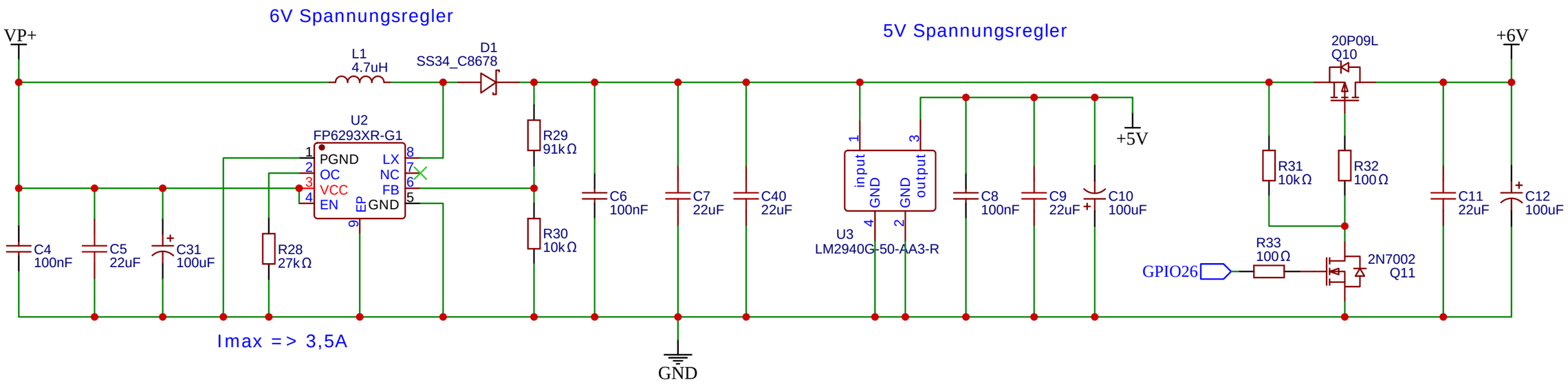
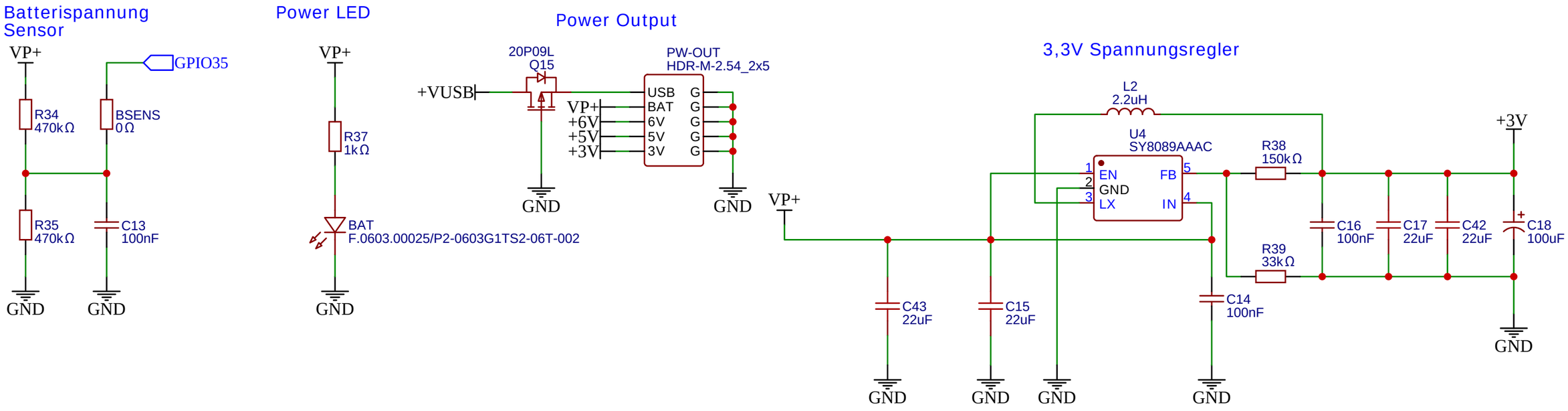

+3V
R1
10kΩ
GPIO00
R2
10kΩ
C1
100nF
U1
ESP32-WROOM-32E
GND
ESP_EN
GPIO36
GPIO39
GPIO34
GPIO35
GPIO32
GPIO33
GPIO25
GPIO26
GPIO27
GPIO14
GPIO12
GPIO13
VDD33
EN
SENSOR_VP
SENSOR_VN
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
IO13
SD2
SD3
CMD
IO23
IO22
TXD0
RXD0
IO21
NC
IO19
IO18
IO5
IO17
IO16
IO4
IO0
IO2
IO15
SD1
SD0
CLK
GPIO23
GPIO22
GPIO01/U0TXD
GPIO03/U0RXD
GPIO21
GPIO19
GPIO18
GPIO05
GPIO17
GPIO16
GPIO04
GPIO00
GPIO02
GPIO15
RST
YTS1C0033BBG01
C2
100nF
R3
10kΩ
R4
10kΩ
Q1
2N7002
R5
10kΩ
R6
10kΩ
Q2
2N7002
+5V
I2C
HDR-M-2.54_1x5
5V
GND
3V
SDA
SCL
R7
10kΩ
R8
10kΩ
Q3
2N7002
R9
10kΩ
R10
10kΩ
Q4
2N7002
R11
10kΩ
R12
10kΩ
Q5
2N7002
R13
10kΩ
R14
10kΩ
Q6
2N7002
SPI
HDR-M-2.54_1x7
MOSI
MISO
SCL
CS0
3V
GND
5V
+6V
SEVO-OPT
HDR-M-2.54_2x4
6V
G
Servoanschl ü sse
HDR-M-2.54_1x3
S
V+
GND
R15
10kΩ
R16
10kΩ
Q7
2N7002
R17
10kΩ
R18
10kΩ
Q8
2N7002
R19
10kΩ
R20
10kΩ
Q9
2N7002
R21
SMD 2 Way Jumper
0Ω
Ausgangsspannung
5V/6V
R36
10kΩ
MODE
YTS1C0033BBG01
Power On LED / Pulldown
R22
1kΩ
R23
10kΩ
EN
GL0603UB01
INFO
XL-0807RGBC-WS2812B
DIN
DOUT
VDD
C3
100nF
LED
R24
1kΩ
R25
1kΩ
R26
1kΩ
R27
10kΩ
ACT
XL-1608UOC-06
Active LED
TITLE:
Steuerung
REV: 4.7
FABLAB
Lübeck
Company: FabLab L ü beck e.V.
Sheet: 1/5
Date: 2026-10-12
Drawn By: Sn@ke





TITLE: Spannungsversorgung		REV: 4.7
FABLAB Lübeck	Company: FabLab Lübeck e.V.	Sheet: 2/5
	Date: 2026-10-12 Drawn By: Sn@ke	

Motorspannung
Extern

MOT-EXT
JL308-25402G01

WJ500V-5.08-2P-14-00A
MOT-EX2

F2
6A

D4
SS34_C8678

GND

+6V

VP+

Motorspannung
Extern/Intern

Intern
6V/Batterie

0Ω
R52

0Ω
R53

Motorspannung
Extern/Intern

Intern
6V/Batterie

0Ω
R54

0Ω
R55

STEPPER-A
XH-4PA

A1 A2 B1 B2

FAULT-R1
LED RED

R40
1kΩ

+3V

U5
AT8833CT_C575404

GPIO26

nSLEEP
AOUT1
AISEN
AOUT2
BOUT2
BISEN
BOUT1
nFAULT

GND
AIN1
AIN2
VINT
GND
VM
VCP
BIN2
BIN1

GPIO16
GPIO25
GPIO32
GPIO27

C22
100nF

C21
10uF

MOT-L
DB128V-5.08-4P-GN-S

M1+
M1-
M2+
M2-

U7
INA2180A1IDGKR

OUT1
IN-1
IN+1
GND
VS
OUT2
IN-2
IN+2

+5V

C28
100nF

R42
200mΩ

R43
200mΩ

C23
10uF

C20
100nF

M4-
M4+
M3-
M3+

DB128V-5.08-4P-GN-S
MOT-R

GND

GND

GND

GND

GND

STEPPER-B
XH-4PA

A1 A2 B1 B2

R46
2.2kΩ

R47
2.2kΩ

FAULT-R
LED RED

R41
1kΩ

+3V

U6
AT8833CT_C575404

GPIO26

nSLEEP
AOUT1
AISEN
AOUT2
BOUT2
BISEN
BOUT1
nFAULT

GND
AIN1
AIN2
VINT
GND
VM
VCP
BIN2
BIN1

GPIO04
GPIO12
GPIO15
GPIO14

C24
100nF

C25
10uF

C26
100nF

C27
10uF

GPIO34
GPIO36

R48
2.2kΩ

R50
2.2kΩ

SENS-C
JUMPER-PAD-2

R44
200mΩ

R45
200mΩ

SENS-D
JUMPER-PAD-2

R49
1kΩ

R51
1kΩ

GND

GND

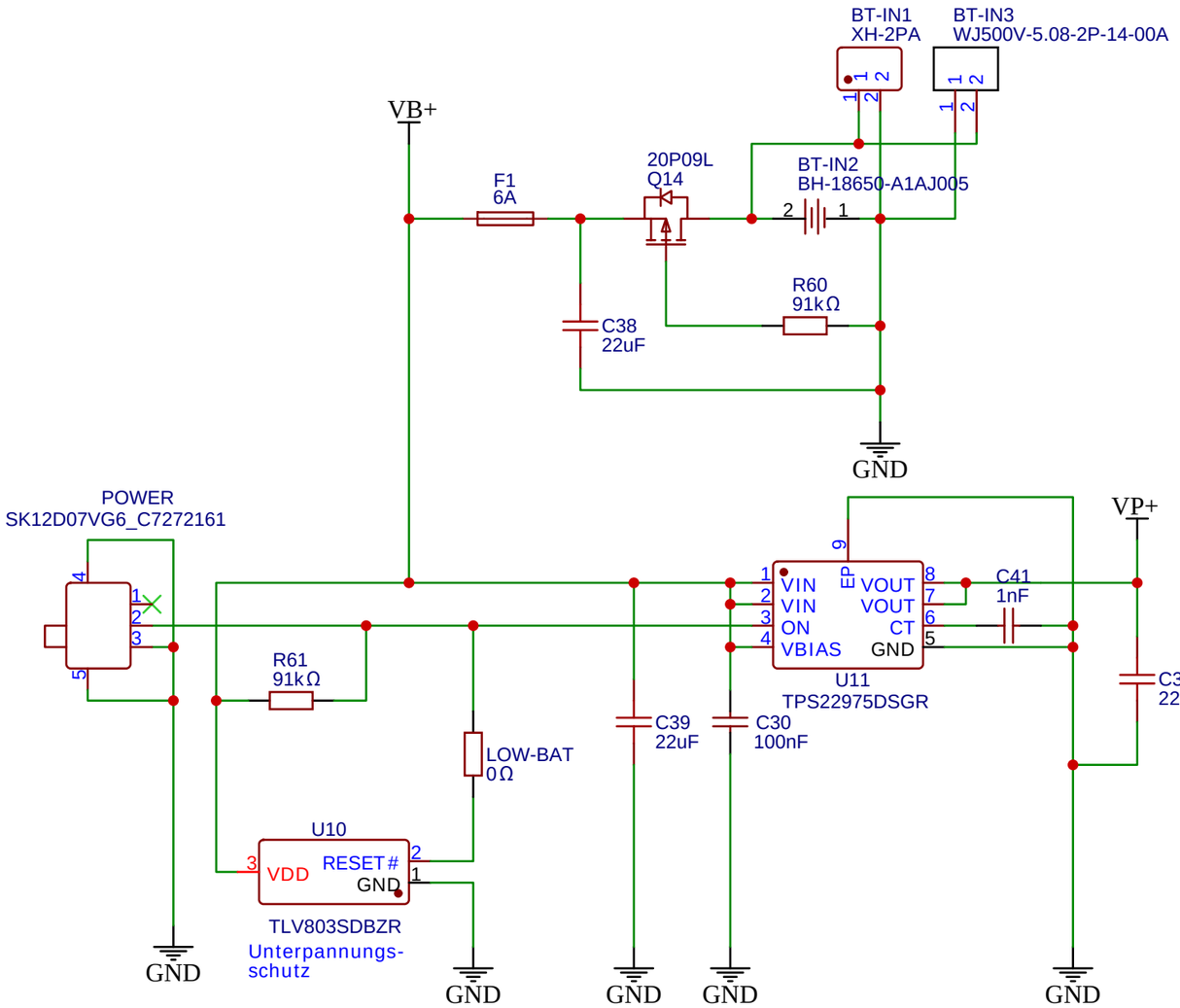
GND

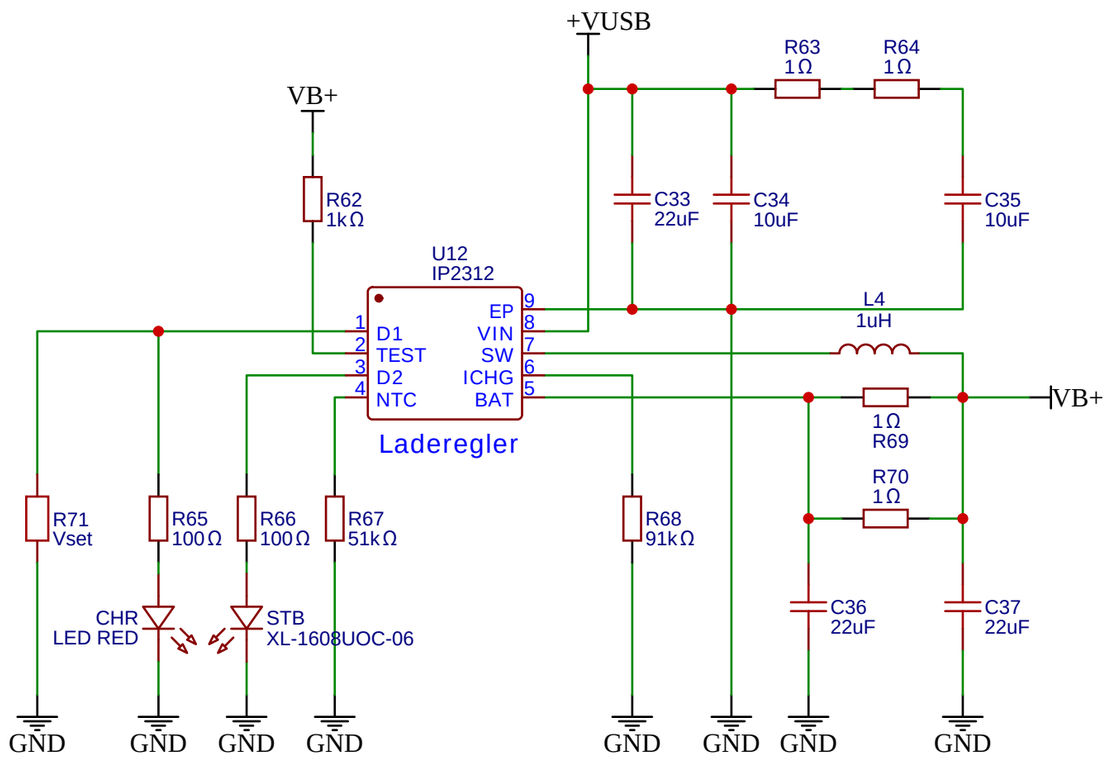
GND

GND

TITLE: Motoransteuerung	REV: 4.7
FABLAB Lübeck	Company: FabLab Lübeck e.V.
	Sheet: 3/5
	Date: 2026-10-12 Drawn By: Sn@ke

TITLE: USB Anschluss		REV: 4.7
FABLAB Lübeck	Company: FabLab Lübeck e.V.	Sheet: 4/5
	Date: 2026-10-12 Drawn By: Sn@ke	





TITLE: Batterie		REV: 4.7
FABLAB Lübeck	Company: FabLab Lübeck e.V.	Sheet: 5/5
	Date: 2026-10-12 Drawn By: Sn@ke	